STAFF REVIEW · H446 075 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. One grows with n; the other does not.

2. Five.

3. Subset backtracking or exhaustive enumeration.

4. 12 elements; last index (3,2).

I Do — source worked steps

1. Explain — Each of the six outer iterations runs six inner iterations.

We Do — guided CHECK

1. Explain
Every new independent binary choice doubles the count.

You Do — source mark schemes

1. n5 $n^2$25 [1] $2^5$32 [1]; n10 $n^2$100 [2] 2^{10}1024 [2].

2. Outer n iterations [2], inner n per outer [2], n^2 total [1], O(n^2) [1].

3. 2^{30}≈1,073,741,824 subsets [2]; impractical exhaustive work may occur [1]; backtracking pruning with valid bound [2], depends on safe assumptions [1].

Exit Ticket — source answer

1. False. It can grow dramatically faster.